

Kevin Crooks Jr.

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EDUCATION

University at Buffalo, The State University of New York

Expected May 2026

Bachelor of Science: Computer Science

Minor: Cybersecurity

GPA: 3.6

Dean's List: Fall 2022, Fall 2023, Spring 2025

Courses: Software Engineering, Systems Security, Computer Security, Systems Programming, Data Structures, Computer Organization, Microprocessors, Artificial Intelligence, Algorithms for Modern Compute Systems, Computer Algorithms & Complexity, Digital Forensics, Cybersecurity, Privacy & Ethics

WORK EXPERIENCE

Undergraduate Teaching Assistant, UB Department of Computer Science and Engineering

September 2023 – present

- Currently a TA for Computer Security, Systems Programming, and Intro to Computer Science
- Hold office hours 3 days a week, teaching labs, and grade homework and exams helping 500 students
- Proctor students in my lab section for lab exams and assist them with debugging their code in lab sections and reinforcing lecture material.

PROJECTS

Spinderfy – Find New Music by Swiping Left/Right on Tracks | React, JavaScript, PHP, GitHub, Scrum

- Used Spotify's API to allow users to curate playlist by swiping left and right on tracks in a similar way to modern dating app but for music
- Developed with a small team with a project manager using agile methods such as creating and implementing user stories and tasks cards into 4 sprints

Memory Allocator Based on Malloc from the C library | C, Emacs, Linux, GitHub, x86-64, MakeFile

- Employed a memory allocation function based on the Malloc/Calloc memory allocators and the memory returned to the user contained metadata which the user can't overwrite and tells the user if it's free
- Represented as a pool allocator where the highest amount of memory is a page size, or 1 kilobyte (4092 bytes) can be allocated to the user, implemented on the x86-64 chip architecture

Web Security Attacks (CSRF, XSS, SQL Injection) | JavaScript, SQL, Virtual Box, Linux, Docker

- Implemented attacks using Seed Labs Virtual Machine with docker containers providing a Social media website to exploit all three attacks.
- Additionally, I implemented the countermeasures for each attack to mitigate the risk of a similar attack

Classic 1972 Atari Video Game Pong | ARM 32bit Assembly, Tiva board, Code Composer Studio

- Presented the game screen on terminal emulator (putty) and ran the project on a Microprocessor board (Tiva) that contained General purpose I/O, UART, and Momentary push buttons
- Customized game play by adding in a power up that allows the player that picked it up to shrink the size of their opponent's paddle while double their own

SKILLS & TOOLS

Languages: Python, C/C++, Java, Scripting (Bash/Powershell), Assembly (ARM/MIPS), Ocaml, HTML, JavaScript, PHP, React, MySQL

Networking/Cybersecurity: MITRE Att&ck framework, TCP/IP attacks, Web Security Attacks (CSRF/XSS/SQL Injection), OSI Model

Conceptual: Basic knowledge of C++/C#/.NET, low-level systems programming/embedded concepts, Object-Oriented Programming (Java, Python, C++), OSI Model

Tools/IDEs: Visual Studio, JetBrains, Wireshark, GitHub, Linux OS, Bash Shell, PowerShell, Microsoft Office

INVOLVEMENT & CERTIFICATIONS

- Achieved 3rd Place at UB Hacking Fall 2025, More detail on my GitHub (Project: FILE_XOR)
- Treasurer & Secretary of Student Excellence in Cybersecurity Club, Spring 2025 – present
- Pursuing CompTIA Security+ & Pentest+ Certifications before May 2026
- Member of the National Society of Black Engineers (NSBE), Fall 2022 – present